CfE2026 — Computer Vision for Ecology

Step 6: Initial Evaluation

Does Temporal Phenological Overlap Matter?

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June 2026

Executive Summary

This report evaluates whether incorporating phenological (temporal) information changes the picture of plant-pollinator interaction prediction compared to using spatial co-occurrence alone. We compute temporal overlap coefficients for 12 of 54 plant-pollinator pairs in the interaction network and test whether the resulting temporal signal is meaningful and robust.

Headline finding: 38.7% of spatially co-occurring plant-pollinator pairs have temporal overlap below 0.3 — substantially exceeding the 15–20% threshold identified as meaningful in the project brief. This result holds across all spatial bin sizes tested (0.5°–3.0°), confirming that temporal modeling is warranted.

1. Background and Research Question

Spatial co-occurrence is the dominant approach for predicting plant-pollinator interactions from observational data. A spatial-only model implicitly assumes that if two species are observed in the same geographic region, they are available to interact. However, this ignores phenological mismatch — a plant may flower in spring while its nominal pollinator is only active in summer, making interaction impossible despite geographic overlap.

This step asks: *among known plant-pollinator pairs that share a geographic bin, what fraction are temporally misaligned? And is the temporal signal robust to the choice of spatial resolution?*

2. Data and Methods

2.1 Data Sources

Analysis uses three datasets: (1) plant flowering events (*plant_flowering_events.parquet*), (2) pollinator observations (*pollinator_observations_v2.csv*), and (3) a plant-pollinator interaction network of 54 known edges. All data is stored on the WashU crow compute server.

2.2 Overlap Coefficient

For each plant-pollinator edge and each 1.5° spatial bin where both species have at least 10 observations, we compute a normalized 52-week activity histogram for each species. The temporal overlap coefficient is the area of intersection between the two distributions:

$$overlap = \Sigma\ min(p_w, q_w)$$

where p_w and q_w are the normalized observation frequencies for the plant and pollinator in week w. This ranges from 0 (no temporal overlap) to 1 (identical seasonal distributions).

2.3 Data Coverage

Metric	Value
Total edges in network	54
Edges with ≥10 obs per species per bin	12 (22%)
Total edge × bin combinations	517
Dominant plant species	Asclepias syriaca (7 of 12 edges)
Spatial bin size (primary)	1.5° × 1.5°
Minimum observations threshold	10 per species per bin

3. Results

3.1 Metric 1 — Distribution of Overlap Scores

The overlap coefficient is broadly distributed across the 517 edge×bin combinations, with mean 0.394 and standard deviation 0.188. The distribution is right-skewed with values ranging from 0.019 to 0.865.

Statistic	Value
Count	517
Mean	0.394
Std	0.188
Min	0.019
25th percentile	0.240
Median (50th)	0.362
75th percentile	0.535
Max	0.865

The median of 0.362 is well below 0.6, the threshold above which the project brief suggests temporal information barely discriminates. This indicates that timing is meaningfully variable across known interactions and cannot be assumed to be near-complete.

3.2 Metric 2 — Headline Result

38.7% of spatially co-occurring edge×bin combinations (200 of 517) have temporal overlap below 0.3. At the edge level, 5 of 12 surviving edges (41.7%) have mean temporal overlap below 0.3. Cumulative threshold analysis shows that 70.4% of combinations fall below 0.5, and 92.6% fall below 0.7.

This far exceeds the 15–20% threshold identified in the project brief as justifying temporal modeling. The implication is that spatial co-occurrence substantially overestimates interaction plausibility: nearly four in ten co-located pairs are seasonally misaligned.

3.3 Metric 3 — Spatial vs. Temporal Divergence

Spatial overlap (Jaccard index of shared bins) and temporal overlap (mean overlap coefficient) show moderate rank correlation (Spearman $\rho = 0.59$, $p = 0.04$) but weak linear correlation (Pearson $r = 0.43$, $p = 0.16$). Five edges in a narrow spatial Jaccard range of 0.31–0.38 show temporal overlap spanning 0.20–0.49 — more than twofold variation in timing alignment for essentially the same degree of geographic co-occurrence.

This confirms that spatial and temporal overlap are related but non-redundant. A spatial-only model would treat these five edges as equivalent; temporal data sharply differentiates them.

3.4 Metric 5 — Sensitivity to Bin Size

The headline metric was recomputed at five spatial resolutions to test whether the result depends on the choice of 1.5° bins.

Bin Size	Edge×Bin Rows	Unique Edges	% Below 0.3	Median Overlap
0.5°	606	11	53.3%	0.288
1.0°	589	12	43.1%	0.331
1.5° (primary)	517	12	38.7%	0.362
2.0°	407	12	37.6%	0.367
3.0°	307	12	34.9%	0.366

The percentage of low-overlap pairs ranges from 34.9% to 53.3% — never approaching the 15–20% threshold below which the signal would be negligible. The monotonic decrease with increasing bin size is consistent with expected smoothing effects: larger bins pool more observations, averaging out temporal variation and artificially inflating overlap estimates. This means 1.5° is a conservative choice and the true mismatch rate at ecologically relevant fine scales is likely higher.

4. Conclusions

- **Timing matters.** 38.7% of spatially co-occurring plant-pollinator pairs are phenologically mismatched (overlap < 0.3), well above the threshold for meaningful signal. Spatial co-occurrence systematically overestimates interaction plausibility.

- **The signal is robust.** The headline result holds across bin sizes 0.5°–3.0°, ranging from 34.9% to 53.3%. It is not an artifact of the chosen spatial resolution.

- **Spatial and temporal overlap are related but non-redundant.** Spearman $\rho = 0.59$ indicates moderate rank correlation, but pairs with nearly identical geographic overlap can differ twofold in temporal alignment. Temporal information adds discriminative signal that spatial methods cannot capture.

- **The case for temporal modeling is strong.** Given data limitations (12 of 54 edges surviving the observation filter), these findings are conservative. The surviving edges are biased toward better-sampled species; the true mismatch rate across all edges may be higher.

5. Data Limitations

- **Sparse edge coverage.** Only 12 of 54 edges (22%) passed the ≥10 observations per species per bin filter. The 42 excluded edges reflect insufficient co-located observational data, not confirmed absence of interaction.

- **Asclepias syriaca dominance.** 7 of 12 surviving edges involve common milkweed, a heavily-observed species. Results may not generalize to rarer plant-pollinator pairs.

- **Small n limits statistical power.** With n = 12 edges, correlation p-values are fragile. Visual patterns and effect sizes are more interpretable than significance tests.

- **Observation effort bias.** Per-species, per-bin normalization absorbs effort within a species but cross-species comparisons may still reflect observer density differences rather than true phenological signal.

- **Temporal overlap ≠ interaction.** Temporal overlap is a necessary but not sufficient condition for pollination. This analysis tests whether timing improves prediction, not whether it guarantees interaction.

6. Recommended Next Steps

The following steps from the project brief remain:

- **Metric 4 — Phenology-aware vs. naive ranking.** For each plant, rank candidate pollinators by (a) spatial co-occurrence count and (b) spatial × temporal overlap. Compute rank correlation to quantify how much temporal information re-orders the top candidates. Requires expanding the analysis to all candidate pairs, not just the 54 known edges.
- **Expanded data coverage.** Investigate whether lowering the minimum observation threshold (e.g., to 5) recovers additional edges without introducing prohibitive noise.
- **Geographic visualization.** Map the spatial distribution of overlap coefficients to identify whether temporal mismatch is geographically structured (e.g., stronger at range margins or in climate-variable regions).
